

CONFIDENTIAL — MANUSCRIPT COORDINATION

COLIDE / CAD-CBA-v1

A briefing for Cheranrach Mahandren, who will lead the paper writing

From Ibteshamul Haque, with Prof. Por as supervisor. This is the first document I am sharing with you. It explains the project, what is frozen, which numbers you may use, how we should work together, and what not to claim. It is **not** the finished paper.

Please read this PDF first (~15–20 minutes). Then pull `github.com/titoatwork/COLIDE (master)` and open `docs/CHERAN_MANUSCRIPT_HANDOFF.md`. Do **not** use `CAD_CBA_v1_MANUSCRIPT.pdf` dated 22 July — it is stale.

Working relationship

Role	Responsibility
You (Cheran)	Lead structure, narrative, venue style, related work, unifying the draft into a paper.
Ibteshamul	Experiments, CUDA/DICC, artifact lookup, number checks, claim hygiene. 24–48 h fact-check.
Prof. Por	Supervisor. High-level updates only — not day-to-day writing.

Pre-manuscript evidence is **closed**. The remaining work is writing — not a new training campaign, and not inventing a post-fix server Block-3 rebench.

1. What the project is

COLIDE is a multi-objective IoT intrusion-detection systems project. The method package is **CAD-CBA-v1**: a CNN–BiLSTM–Attention student, trained with focal loss and ensemble knowledge distillation from tree teachers (RF + XGBoost + LightGBM), under a sealed BoT-IoT protocol. Alongside detection we measure efficiency (latency, energy, memory) and custom CUDA kernels for the CNN–BiLSTM blocks only (**Option A** — not a full-model CUDA clone).

The scientific story is **honest accuracy–efficiency**, not “we beat every baseline on F1.” Protocol LightGBM still leads pure validation F1. Trees remain strong on tabular flows. Our contribution is a frozen, protocol-fair package plus operation-matched CUDA and multi-GPU measurement — with explicit non-claims.

Working title (locked for now)

CAD-CBA: A Class-Aware Distilled CNN–BiLSTM for Multi-Objective IoT Intrusion Detection with Operation-Matched CUDA Acceleration

Authors / affiliations / venue are for you and Prof. Por to finalise.

2. Current state (15 August 2026)

Layer	State
Pre-manuscript evidence pack	CLOSED — numbers, claims, tables, figures, handoff frozen.
Champion model	Frozen. md5 80a90f7cc210276300eaa90173a5a385. Do not retrain.
ToN-IoT leakage issue	Closed. New leakage-safe run is the only active ToN result.
CUDA Block-3 correctness (laptop)	Closed locally: race/align fix, sanitizers, real-weight parity.
CUDA Block-3 latency on DICC servers	Historical pre-fix only (Option B). No post-fix campaign.

Layer	State
GitHub public repo	Polished. Stale files kept and labeled, not deleted.
Manuscript Markdown	Working draft with synced numbers — feedstock for you to rewrite.
July 22 manuscript PDF	STALE. Please ignore for writing.
Camera-ready paper	Not started. That is your phase.

3. The paper's questions (locked answers)

These research questions are already answered by artifacts. Your job is to write them clearly, not to reopen the experiments unless something is scientifically wrong.

RQ	Locked answer
Detection (BoT sealed test)	Macro-F1 0.9780 ± 0.0033 (n=5). Near protocol RF; does not beat protocol LGBM.
Minority (Theft)	Yes on sealed test: Theft F1 mean 1.0; min-class mean 0.9292.
Local efficiency (RTX 3050)	Exploratory energy 0.920–0.943 mJ/flow; PT@256 ~24.2–25.7 μ s; VRAM 322 MiB.
Cross-GPU / multi-day	DICC tables exist. B3 server timing is pre-fix historical; PT wins B3 wall-clock.
Explainability	Dispatch 16.60 μ s p99 only. Not a free-form LLM-XAI paper.
Second dataset (ToN)	Leakage-safe CNN 0.8075 vs RF 0.9626. Clean 0.9526 path is INVALID.

4. Locked numbers you may put in the paper

4.1 Principal detection — BoT-IoT

Item	Use this
Principal test macro-F1	0.9780 ± 0.0033 (sealed multi-seed, n=5)
Champion	model/best_model_botiot_twostage.pth
MD5	80a90f7cc210276300eaa90173a5a385
Protocol LGBM val (pure F1 leader)	0.9818 — we do not claim to beat this
Protocol RF val	0.9778
Published RF 0.9864	Different pipeline — dual-bar only, not our protocol
Historical 0.9790	Development/legacy only — never as principal

4.2 ToN-IoT — secondary, leakage-safe

Item	Use this
Protocol	toniot_leakage_safe_v1 (13 features, split first, no SMOTE, no KD)
CNN test macro-F1	0.8075
RF test macro-F1	0.9626 (same split)
Split	Seed 42, 60/20/20 stratified random — not official temporal/host split
Weak class	mitm: F1 ≈ 0.111, precision ≈ 0.059, recall ≈ 0.91
Never use as valid	Clean CNN 0.9526 / RF 0.9851 / “+15.4%” (label leakage)

4.3 Systems / CUDA — wording is as important as the number

Item	How to write it
Local B3 parity	Closed. GPU vs PyTorch full sequence max $ \Delta \approx 3.43 \times 10$ ■■■.
Local sanitizers	racecheck / synccheck / initcheck / memcheck: 0 errors (RTX 3050).
DICC B3 latency	Historical pre-fix only. Example: V100S ~513 μ s CUDA vs ~363 μ s PT.
Custom CUDA vs full V3	Forbidden as a model-level speedup. Option A = Blocks 1–4 only.
Framework tables	Separate operator table vs full-model table. No cross-table speedups.
~25,899 flows/s	Bulk batched throughput (batch 128). Not a streaming SLA.
Energy 0.920–0.943 mJ/flow	Exploratory GPU-board energy (WP6b ranges).
16.60 μ s p99	Alert-construction / queue dispatch only.
Native TensorRT equivalence	Not checked — do not claim numerical parity.

If a sentence would be stronger than this table, stop and ask me before it goes into the draft. I would rather weaken a claim than reopen an invalid number.

5. Figures (from locked artifacts)

These are the current figures on disk. Captions below are honest. You may redraw for venue style; do not change the underlying numbers.

5.1 Architecture (schematic)

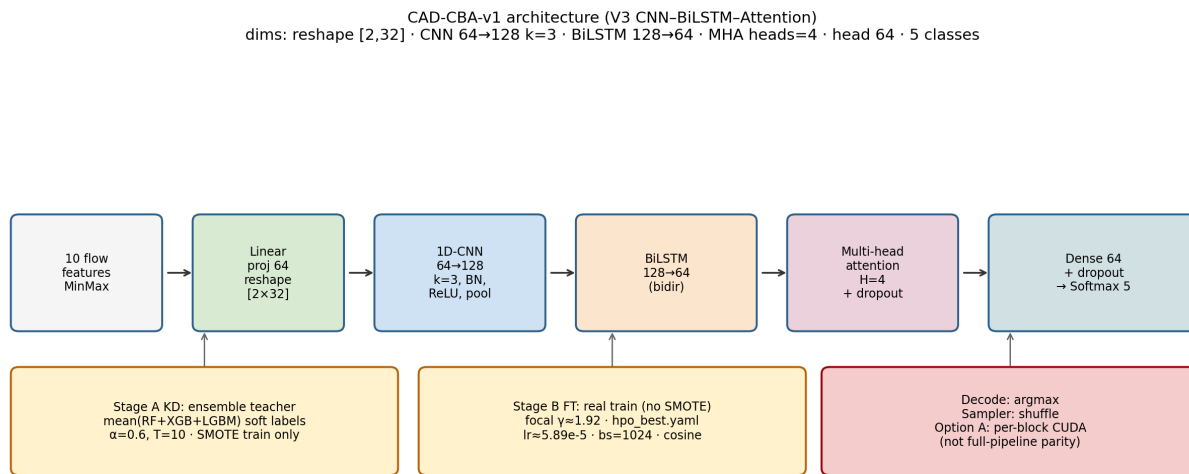


Figure 1. CAD-CBA-v1 schematic (Option A CUDA covers fused Blocks 1–4 only; attention / LayerNorm / residual / classifier remain in PyTorch). Source: docs/manuscript/figures/fig_architecture.png.

5.2 Detection dual bars (protocol honesty)

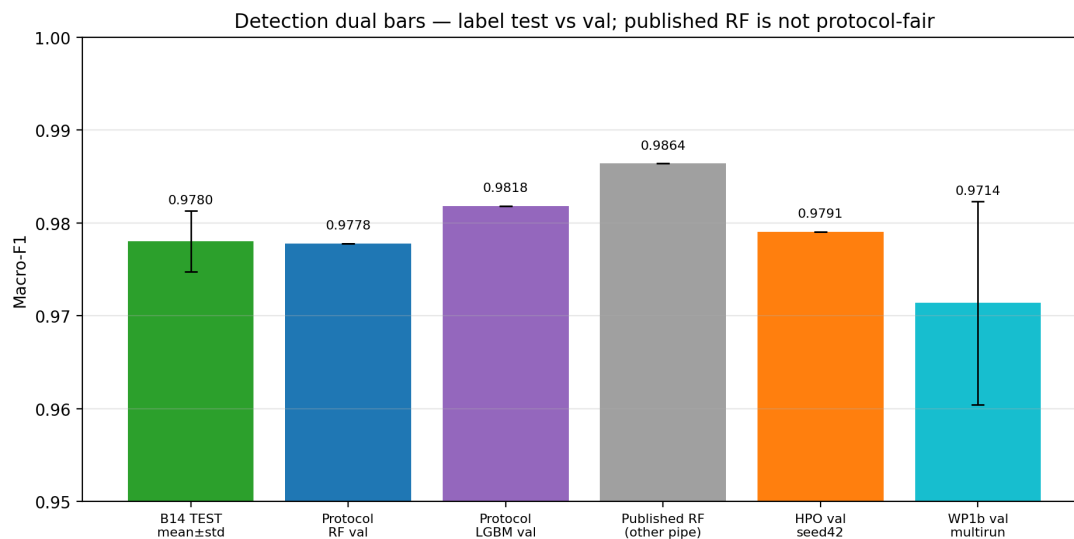


Figure 2. Dual-bar detection view: sealed CAD-CBA test versus protocol and published baselines. Do not collapse published RF 0.9864 into our protocol. Source: fig_detection_dualBars.png.

5.3 Sealed-test confusion (BoT, seed 42 representative)

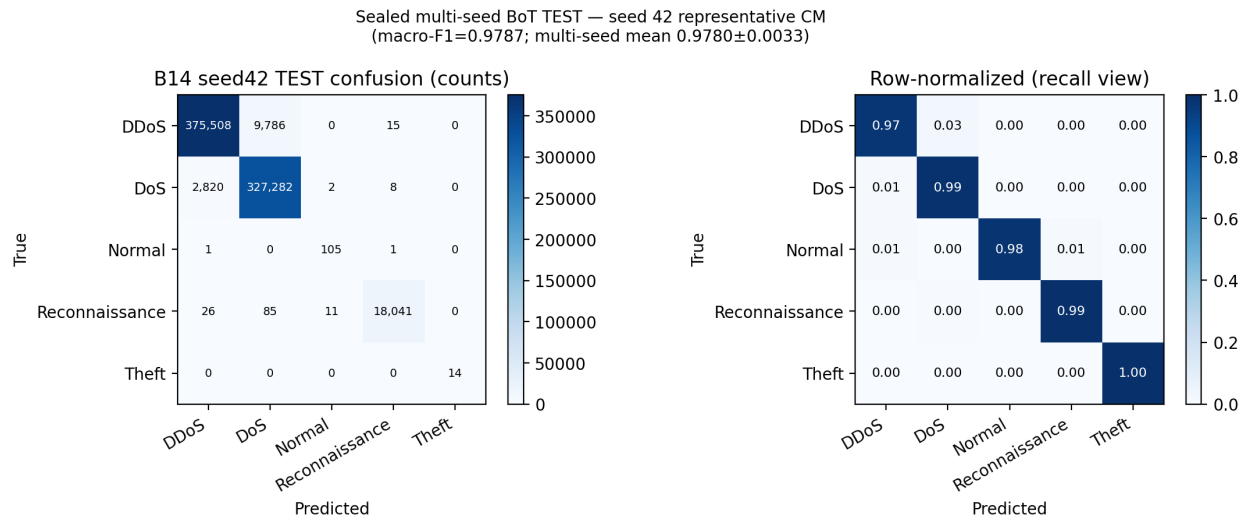


Figure 3. Sealed BoT test confusion, seed 42 representative (seed-42 F1 $\approx$ 0.9787; multi-seed mean 0.9780±0.0033). Source: fig_confusion_matrix_b14_seed42.png.

5.4 Ablation ladder (package credit, not attention-alone)

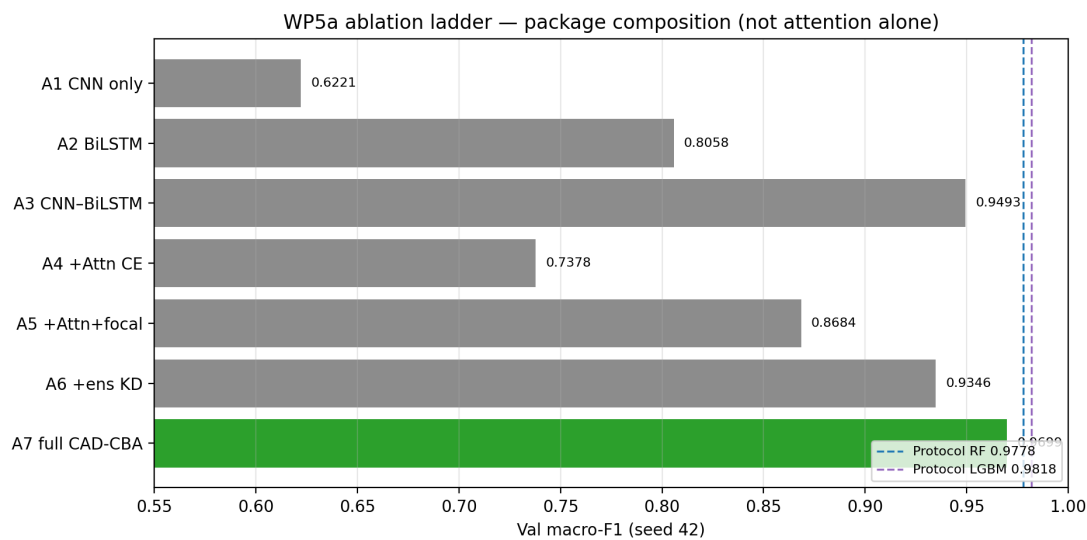


Figure 4. Ablation ladder. Attention+CE alone is a negative result versus CNN-BiLSTM; credit the frozen package. Source: fig_ablation_ladder.png.

5.5 Corrected ToN per-class (CNN)

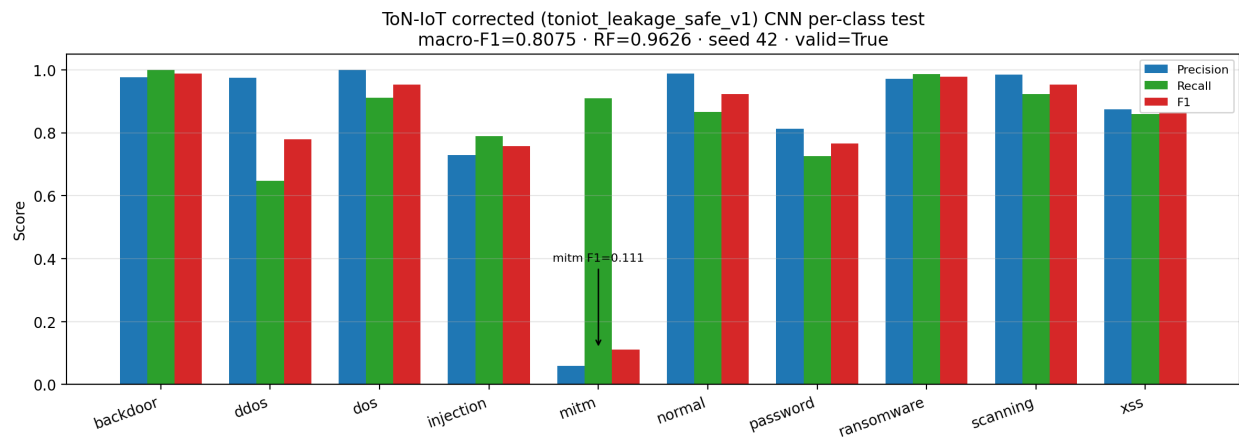


Figure 5. ToN leakage-safe CNN per-class test. Macro-F1 0.8075 vs RF 0.9626. mitm is weak ($F1 \approx 0.11$). Do not replace this with the invalid “clean” 0.9526 table. Source: fig_toniot_corrected_cnn_per_class.png.

5.6 Local systems ranges (exploratory energy / latency)

Local systems ranges — RTX 3050 laptop only (not DICC multi-GPU)

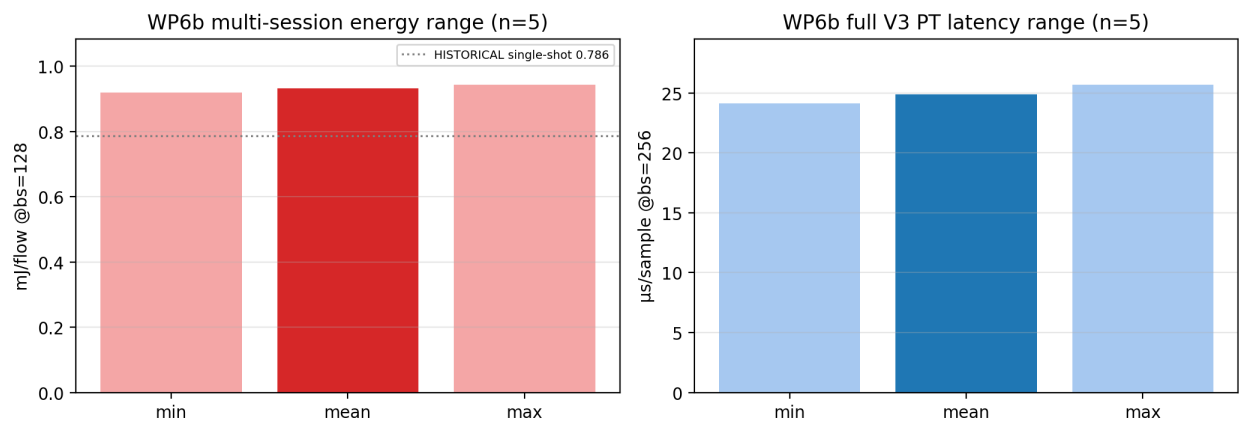


Figure 6. Local multi-session WP6b ranges on RTX 3050 (exploratory energy; not DICC). Source: fig_wp6b_systems_ranges.png.

6. Hard non-claims (please keep these in the paper)

- We do not claim pure-F1 supremacy over protocol LightGBM.
- We do not claim full custom-CUDA versus full V3 PyTorch speedup or parity.
- We do not claim DICC Block-3 latency as post-fix, or that CUDA B3 beats cuDNN on servers.
- We do not claim production LLM explainability.
- We do not cite ToN “clean” 0.9526 / 0.9851 / +15.4% as valid.
- We do not call ~25,899 f/s a live streaming SLA, or energy a certified metrology result.
- We do not treat the 32 latent steps as observed packet chronology (pseudo-sequence).

7. How I suggest we work

7.1 Suggested division

You draft	I support
Title/abstract variants, intro, related work, discussion, conclusion	Artifact lookup, CUDA/DICC wording
Unify MD → Word or LaTeX for the venue	Regenerate a table/figure from JSON if needed
Related-work citations and flow	Confirm any new quantitative sentence against JSON
Flag any claim that feels too strong	Reply with the allowed wording or a weaker sentence

7.2 Source of truth (in this order)

1. Gate JSON under benchmarks/results/ (ToN, B3 parity, framework, sanitizer).
2. docs/RESULTS_INDEX.md and docs/CLAIM_MAP_PREWRITE.md.
3. docs/manuscript/TABLES_FROM_ARTIFACTS.md and docs/manuscript/figures/.
4. Working draft docs/manuscript/CAD_CBA_v1_MANUSCRIPT.md — rewrite freely; keep numbers.
5. This briefing — orientation only.

Older emails, HANDOFF.md, STATUS_REPORT_DRAFT, and PROF_POR_* packs are historical. If they conflict with RESULTS_INDEX, ignore them.

7.3 What to open after this PDF

Order	File
1	docs/CHERAN_MANUSCRIPT_HANDOFF.md
2	docs/CLAIM_MAP_PREWRITE.md
3	docs/RESULTS_INDEX.md
4	docs/manuscript/CAD_CBA_v1_MANUSCRIPT.md (working draft)
5	docs/manuscript/TABLES_FROM_ARTIFACTS.md
6	docs/B3_SERVER_LATENCY_DECISION.md (Option B)
7	docs/KNOWN_LIMITATIONS.md

8. Access

Repository (preferred): <https://github.com/titoatwork/COLIDE> — branch **master**. Public academic repo (Academic Research license, not MIT).

Optional zip (~2.1 MB, writing + figures + key JSON): COLIDE_Cheran_manuscript_pack_20260815.zip — I can re-send with this PDF.

9. A note on tone

This project went through a serious remediation: we withdrew an invalid ToN result, fixed Block-3 kernel bugs, and refused to invent a server rebench we did not run. The paper should sound like that — confident where the artifact is strong, and explicitly modest where it is not. That honesty is part of the contribution.

Thank you for taking the manuscript lead. I will stay available for numbers, CUDA language, and DICC tables whenever you need a check.

Next 48 hours (suggested)

1. Reply with preferred venue and Word vs LaTeX.
2. Clone the repo (or open the zip) and skim the seven files in §7.3.
3. Draft a one-page outline (sections + which figures). I will sanity-check claims.
4. Ignore the 22 July manuscript PDF entirely.

Ibteshamul Haque · FCSIT, Universiti Malaya · COLIDE under Prof. Por · 15 August 2026. Authority chain: JSON → RESULTS_INDEX / CLAIM_MAP → manuscript MD → this briefing.